# Quantum-Chemistry Assistant: Local Adaptation Workflow

Public references + original exercises | Apple Silicon | Fixed diagnostic prompts

### 1 Curate sources

Textbook concepts and papers
PySCF API and public code
Record version + provenance

### 2 Build exercises

Original bilingual Q&A
88 train / 12 validation
8 fixed diagnostic prompts

### 3 Validate data

Check formulas and code
Check exact prompt overlap
Check chat token lengths

### 4 Establish baseline

Pinned MLX model revision
Greedy decoding, seed 42
Save responses + stop reasons

### 5 Train low-rank adapters

Frozen 4-bit base weights; group size 64
Last 4 blocks: query + value projections
Rank 8; MLX scale 16; Adam; learning rate 0.0001
Batch 1; 176 steps; answer-token loss only
Validation every 44 steps; gradient checkpointing

### 6 Compare responses

Same model and tokenizer
Same prompts and decoding
Review correctness and code
Report repetition + truncation
Use the fixed final checkpoint

### 7 Package the study

Adapter + reproducible scripts
Data and public references
English report and manuscript
Publish only reviewed files

## Inside each adapted projection

**Frozen branch**

x -> quantized linear layer

+

**Trainable branch**

x -> A -> B -> multiply by scale

**Output**

Sum both branches

**Supervision**

Predict answer tokens

$y = Q(W0)x + sBAx$ | Only A and B are updated; source documents are not copied into the training set.